Introduction to Vector Motion

Scalars and vectors

Scalar: Is a value that has _____ (will also include units).

Vector: Is a value that has a magnitude and _____.

To show that a variable is a vector, we draw _____.

Ex. Velocity is a vector: The variable used for velocity is _____

When we want to specify the direction of a vector, typically we first write the _____ (with units) and then the direction in _____.

Ex. Sarah is driving north in her Tesla Model S at 175 km/hr. Express this as a vector: _____

Other important scalars and vectors:

Scalars	Vectors

Distance, Position and Displacement

Distance (_____): The length of _____.

Position (_____): Is an object's _____.

Typically, position is the _____ _____ along with a direction to that location, measure from the reference location.

Displacement (_____): Is the _____.

Displacement can be thought of as the straight line between an object's location at the start of an interval, with a direction (straight from the interval start to the interval end).

Think / Pair / Share:

Question 1: When is displacement equal to distance? _____

Question 2: When is displacement greater than distance? _____

Example 1A: Mrs. Yang is standing 1 m west of her reference position. She then:

A. Walks 1 m east

B. Walks 1 m east

C. Walks 2 m east, then turns and walks 3 m west

Fill in the following table:

Interval	Position at start of interval	Position at end of interval	Interval displacement	Total displacement
A				
B				
C				

*Note: Reference position is defined as 0 m.

Often to make life easier, we define a specific direction (or directions) as being _____. This means that the _____. These are called "direction conventions".

Example 1B: Same as example 1A except set East as the positive direction (means that west is negative).

Interval	Position at start of interval	Position at end of interval	Interval displacement	Total displacement
A				
B				
C				

*Note: End of each interval is the start of the next interval.

<u>Displacement Formulas</u>

$\vec{\Delta d} = \vec{d}_f - \vec{d}_i$

Meaning that displacement = _____

$\vec{\Delta d}_{1-3} = \vec{\Delta d}_{1-2} + \vec{\Delta d}_{2-3}$

Meaning that if the object is moving in a series of displacements, then total displacement is

_____

Example 2: Bob starts 10 km north of the school. He then travels 20 km south, then 12 km north. (Assume the school is the reference location).

Determine (in any order) Bob's initial position (_____), final position (____), total distance (_____), and total displacement (____). Include a labelled sketch.

Example 3: If Sean walks 2.0 km [E], then 4.0 km [S], what is his displacement?

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SPH3U PHYSICS

Vector Motion Problems: Distance, position and displacement

1. Draw a sketch for each situation. Then determine the initial position, final position, total displacement and total distance for each of the following. Show your work.
   a. Starting from home Val travels 9.0 m [E], then 14.0 m [W], and finally 8.0 m [E].

   b. Therese travels from a position 5.0 km [N] of her home to a position 3.0 km [S] of it.

   c. Starting from home, Robbie travels 3.0 m [N], then 5.0 m [S], and finally 4.0 m [S].

   d. Starting at a position of 4.0 m [E] of home, Sangar travels 10.0 m [W], then 15.0m [E] and finally 8.0 m [W].

e. Starting from school, Tim travels 1.0 km [W], 2.0 km [N] then 4.0 km [W].

f. Starting 2.0 km [N] of the street, Mike runs 5.0 km [S], 4.0 km [W], 2.0 km [N], 1.0 km [E], 3.0 km [N] then 3.0 km [E].

Other Questions:

Is the final position always the same as the total displacement of the object? What must be the case when they are the same?

When is the total displacement the same as the total distance?

SPH3U Name: _____

Average Speed vs Average Velocity

Average Speed = (distance travelled) / (time taken)

$$v_{ave} = \frac{d}{\Delta t}$$

Average Velocity = (displacement) /(time taken)

$$\vec{v}_{ave} = \frac{\Delta \vec{d}}{\Delta t}$$

Questions.

1. When is distance equal to displacement?

2. Mrs. Yang ran a "10 K" charity race. She ran the first 5.0 kilometres in 36 min 56 s. Then she ran the second 5.0 kilometres in 35 min 40 s.

a) Calculate her average velocity, if the race starting point was the same as the finish point.

b) Calculate her average speed in m/s.

3. Using the speed of light in a vacuum (3.0 x 10^8 m/s) and the average distance from the Sun to Earth (150 million km), find how many minutes it takes for sunlight to reach Earth.

Describing Motion Verbally with Speed and Velocity

Read from Lesson 1 of the 1-D Kinematics chapter at The Physics Classroom:

http://www.physicsclassroom.com/Class/1DKin/U1L1d.html

MOP Connection: Kinematic Concepts: sublevels 3 and 6

Review:

1. A _____ quantity is completely described by magnitude alone. A _____ quantity is completely described by a magnitude with a direction.
   a. scalar, vector b. vector, scalar
2. Speed is a _____ quantity and velocity is a _____ quantity.
   a. scalar, vector b. vector, scalar

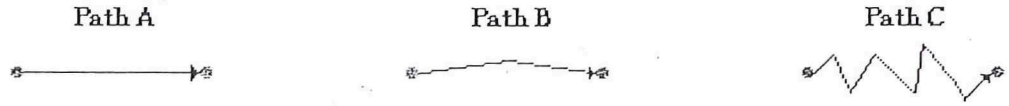
Speed vs. Velocity

Speed and velocity are two quantities in Physics which seem at first glance to have the same meaning. While related, they have distinctly different definitions. Knowing their definitions is critical to understanding the difference between them.

Speed is a quantity which describes how fast or how slow an object is moving.

Velocity is a quantity which is defined as the rate at which an object's position changes.

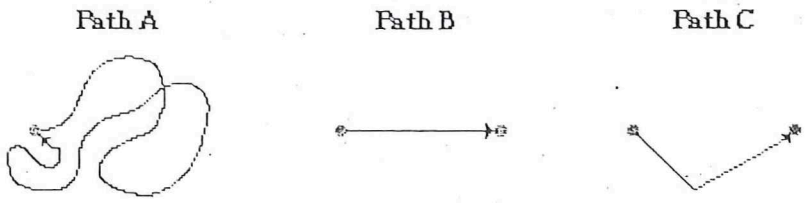
3. Suppose you are considering three different paths (A, B and C) between the same two locations.



Along which path would you have to move with the greatest speed to arrive at the destination in the same amount of time? _____ Explain.

4. True or False: It is possible for an object to move for 10 seconds at a high speed and end up with no overall change in position.
   a. True b. False
5. If the above statement is true, then describe an example of such a motion. If the above statement is false, then explain why it is false.

6. Suppose that you run for 10 seconds along three different paths.



Rank the three paths from the lowest speed to the greatest speed. _____

Rank the three paths from the lowest velocity to the greatest velocity . _____

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DT Graphs, Speed and Velocity

1. Follow the instructions provided by your teacher to collect the data.

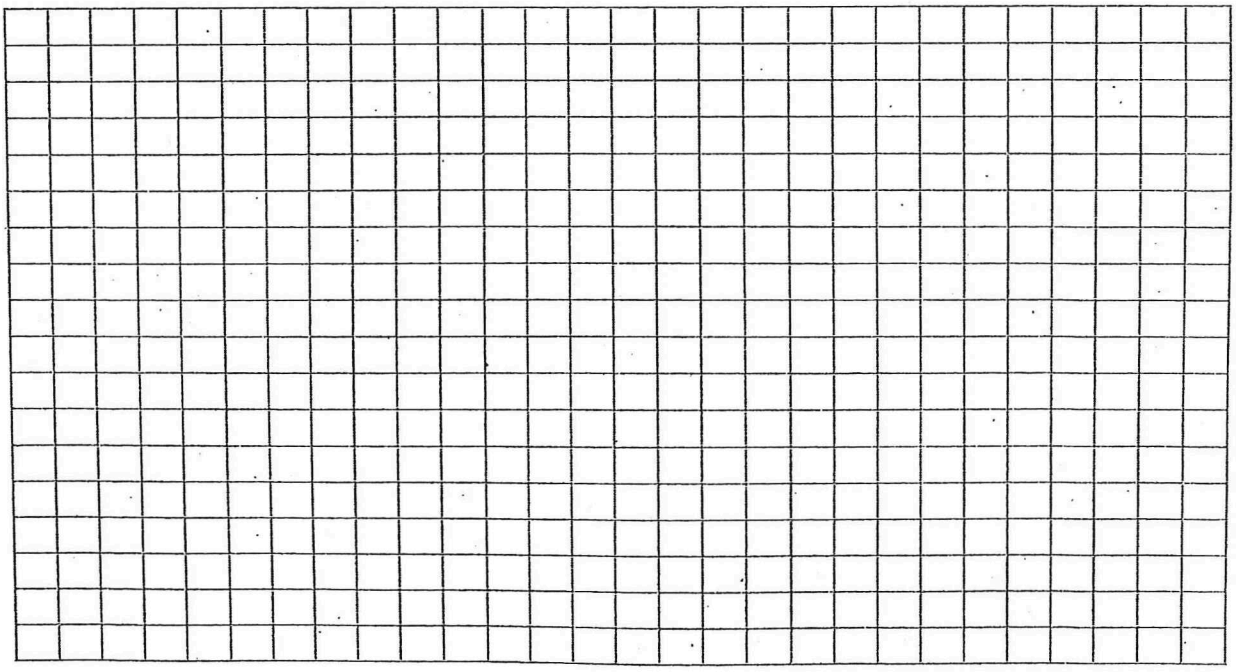
Experiment 1: Ball rolling slowly

Time (s)	Position (cm) [ ]

Experiment 2: Ball rolling quickly

Time (s)	Position (cm) [ ]

2. Plot both sets of data on the same graph below.
   - Always include a proper title and labels on your graphs.
   - Make sure time is on the x-axis and position is on the y-axis.
   - Include a line of best fit for each experiment. Use different colours and label each line.



3. Calculate the slope of each line of best fit.

Slope for experiment 1: _____ Slope for experiment 2: _____

Analyse

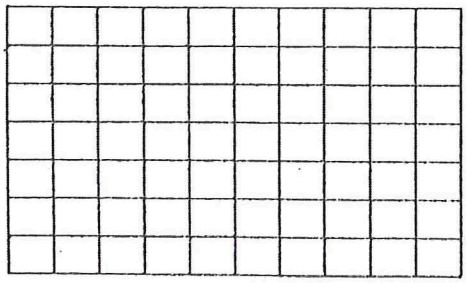
1. What is the reference point you used to measure position? _____
2. Is the position you measured a scalar or a vector? _____
3. Are the slopes you calculated scalars or vectors? _____
4. What does the slope represent? _____
5. What is the speed of each experiment? _____ _____
6. What is the velocity of each experiment? _____ _____

Extend

The graph you created is called a DT graph.

1. Roughly sketch what you would expect a DT graph to look like in the following situations:
   a. A rock is not moving.
   b. A ball is rolling backwards.

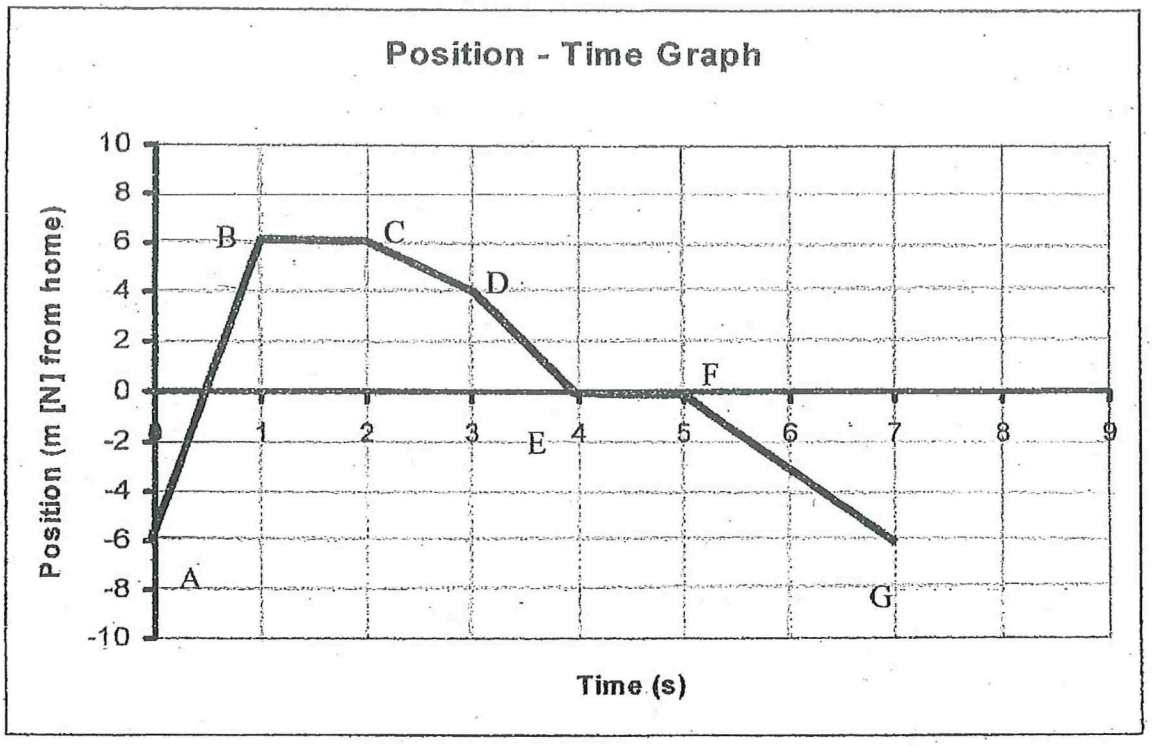
2. A ball is rolling forwards at 1.0 m/s for 6.0 seconds, but then is kicked backwards at 2 m/s back to where it started.
   a. Graph the situation on the grid provided.
   b. Calculate the average speed of the ball between 0 s and 9.0 s.
   c. Calculate the average velocity of the ball between 0 s and 9.0 s.



3. Is it possible for your line of best fit on a DT graph to be vertical? Explain why or why not.

Position-Time Graphs

1. The position - time graph for an object in motion is shown below.



Using the above graph determine the following.

c) What is the displacement in the intervals:

i) AC _____ ii) DG _____

b) What is the distance traveled in the intervals:

i) AC _____ ii) DG _____

c) What is the velocity in the intervals:

i) AB _____ ii) FG _____

c) What segment of the trip has a zero velocity? Explain?

e) What is the average speed in the intervals:

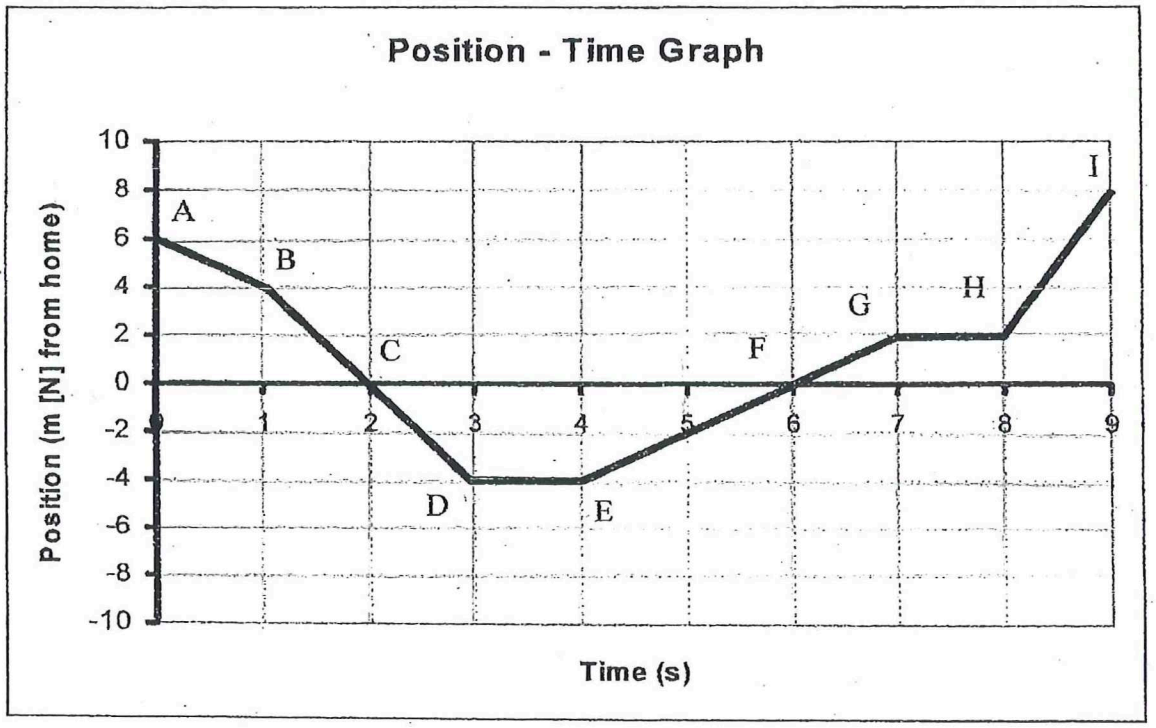
i) AC _____ ii) DG _____

iii) AG _____

f) What is the average velocity in the intervals:

i) AC _____ ii) DG _____

2. The position - time graph for an object in motion is shown below.



Using the above graph determine the following.

d) What is the displacement in the intervals:

i) AD _____ ii) EI _____

b) What is the distance traveled in the intervals:

i) AE _____ ii) DH _____

c) What is the velocity in the intervals:

i) BC _____ ii) FG _____

d) What segments of the trip have a zero velocity? Explain?

e) What is the average speed in the intervals:

i) AE _____ ii) DH _____

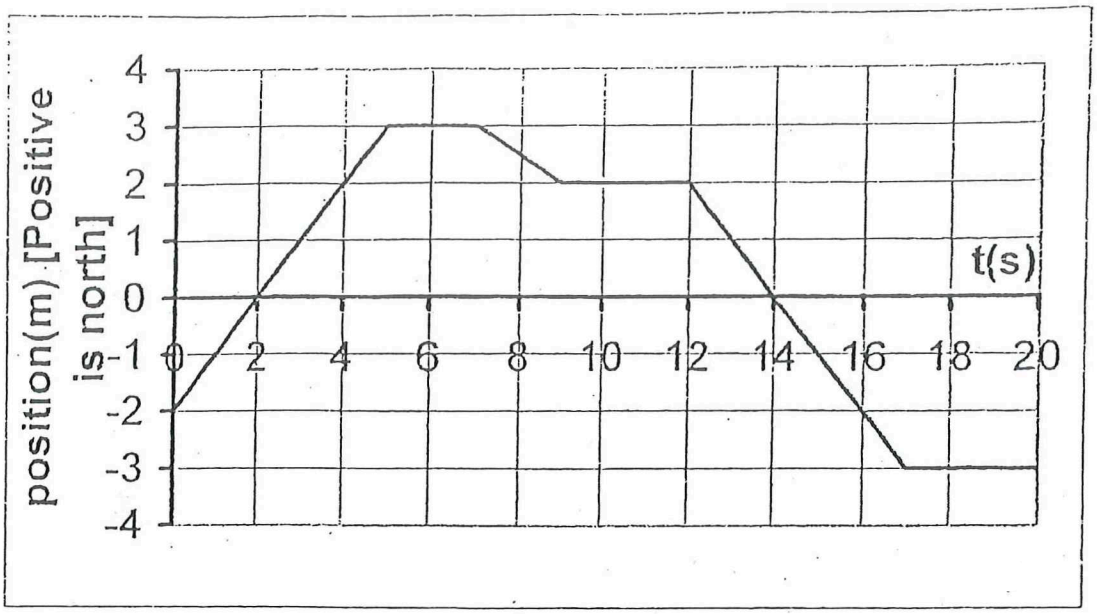
iii) AH _____

f) What is the average velocity in the intervals:

i) AC _____ ii) DH _____

Position-time Graphs

Consider the graph below to answer all of the questions below, showing all of your work.



1. At what time(s) is the object at rest?
2. What is the velocity of the object at t = 4.36 seconds?
3. The graph does not start at position zero. Explain what this means.
4. At what time(s) is the object north of the starting point?
5. At what time(s) is the object north of the reference point?
6. What is the average velocity for the entire 20 seconds?
7. At what time(s) is the speed of the object a maximum?
8. What is the displacement of the object for the entire 20 seconds?
9. At what time(s) is the object travelling north?
10. At what time(s) is the object travelling south
11. At what time(s) is the object travelling east?
12. What is the displacement between t = 4 seconds and t = 16 seconds
13. What distance does the object travel during the entire 20 seconds?
14. What is the average speed for the entire 20 seconds?
15. At what time(s) is the object 2 m [N] of the reference point?
16. What does a negative slope mean on a position-time graph?
17. What does a positive slope mean on a position-time graph
18. What should we look for on a position-time graph if the velocity changes?
19. Does the velocity change on this position-time graph? If the answer is yes, at what time(s) does it change?

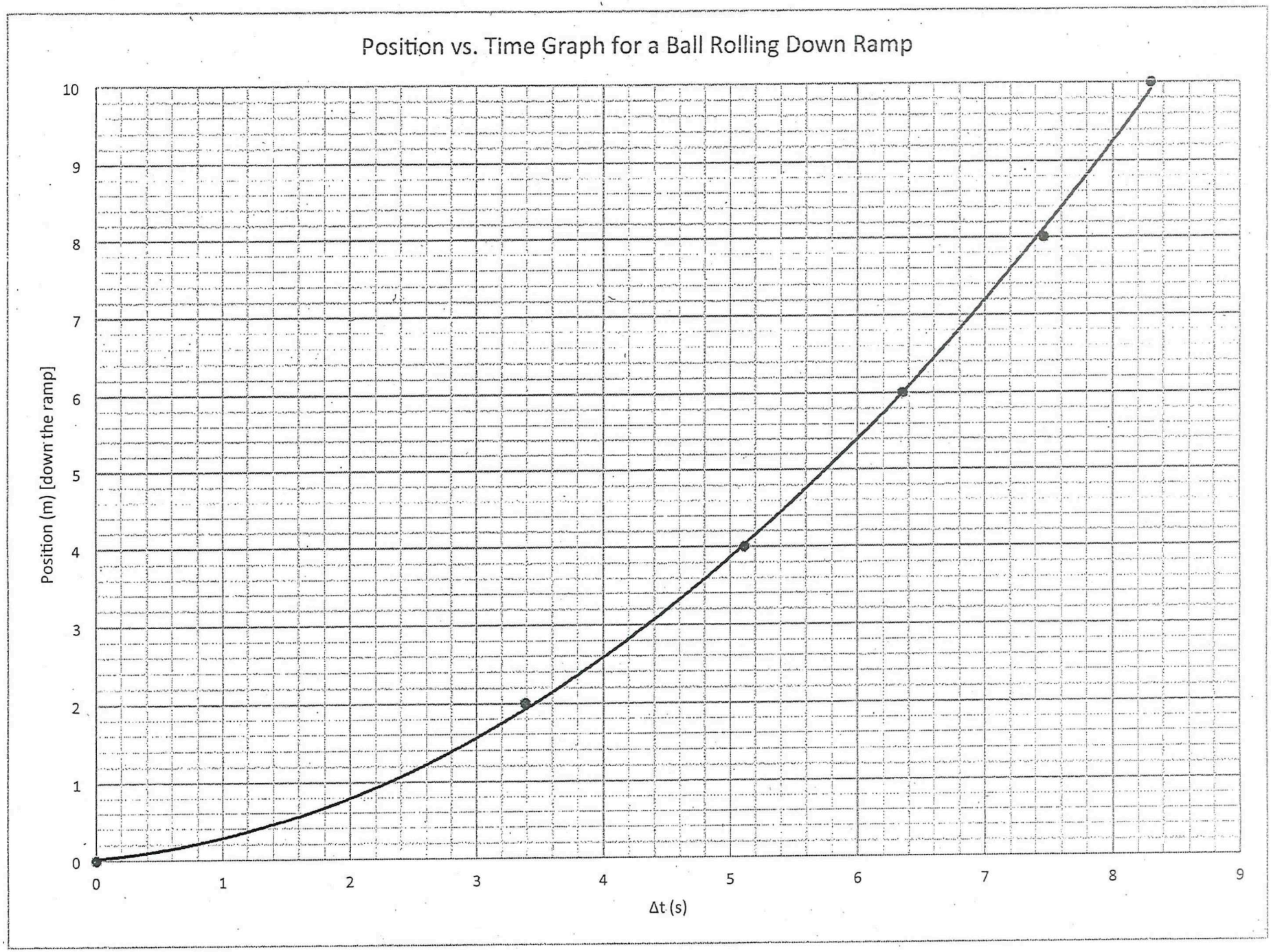
Kinematics Lab #2

Sledding down a hill.

1. Based on your data table, plot the set of ordered pairs on a position vs. time graph (this is graph #1).
   NOTE: Make your graph take up the entire page – a larger graph leads to less error when drawing your tangent lines!
   In addition, it is easiest to draw your graph in "landscape": 
   rather than "portrait". 
2. Draw a curve of best fit for the points.
3. How can you tell, by looking at the curve, that the sled is speeding up, as it goes down the hill? Explain your answer, with reference to kinematics lab #1.
4. Draw at least 5 tangents to the graph. Try to make your tangent lines evenly spaced out. Make sure that the tangent lines are reasonably long.
5. Find the slope of each of the tangent lines. Include units. Show your work.
6. What does the slope of each of the tangent lines represent?
7. Make a new table of values:

Time at which tangent touches the graph	Slope of tangent

8. Draw a new graph (graph #2), plot the ordered pairs from your table of values on this new graph.
9. Draw a line of best fit for this data set. Find the slope of the line of best fit. Include units. Show your work.
10. Draw a new graph (graph #3), plot the slope of the line of best fit from graph #2, at various times.
11. Draw a line through the points on your graph.
12. Give realistic sources of error for this lab. For each source of error, suggest plausible changes that could be made, to reduce this error.


Position vs. Time Graph for a Ball Rolling Down Ramp
Position (m) [down the ramp]
0
1
2
3
4
5
6
7
8
9
10
Δt (s)
0
1
2
3
4
5
6
7
8
9